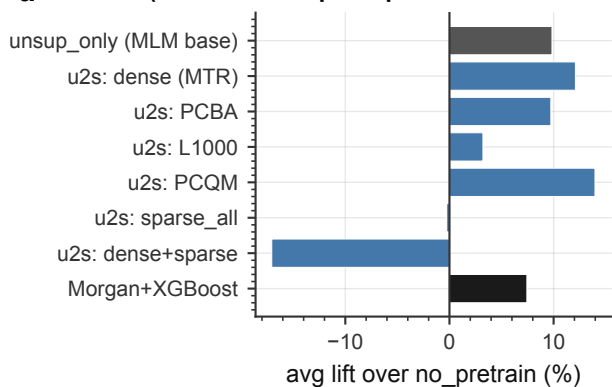
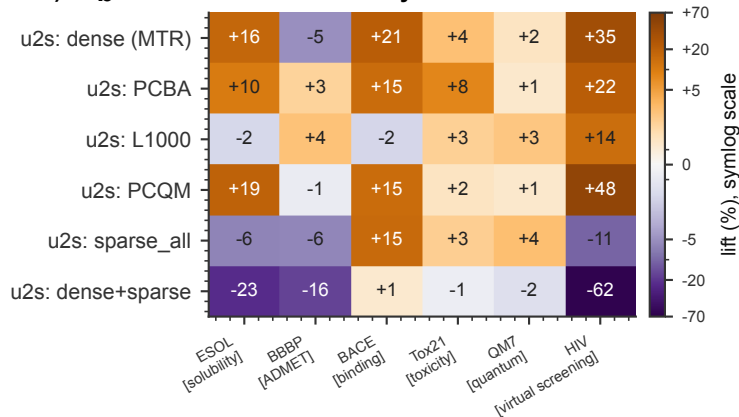


Which SFT pretrain helps? Which type helps, how much, and does it follow chemistry?

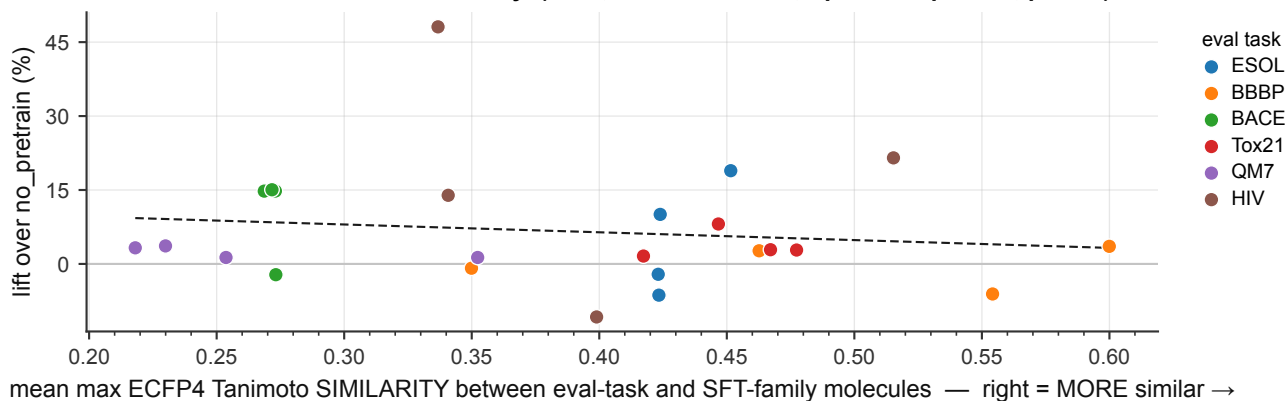
**a** (all arms: unsup→sup from one 2M-FP MLM base)



**b** SFT family → eval task



**c** H10 test: lift vs chemical similarity (n=24, Pearson  $r=-0.14$ , Spearman  $\rho=-0.10$ ,  $p=0.64$ )



deduped wave: the SFT blocklist drops 34,301 eval molecules, so no arm trains on eval-test molecules. Arms are unsup→sup from one shared 2M-FP MLM base, each lifted against the random-init floor scored in its own wave.

SCHEME: single DeepChem scaffold hold-out (this wave has no 5-fold CV eval), so these lifts are NOT comparable with the CV numbers in A1.b / A2.

(c) omits dense (MTR) and dense+sparse: descriptor regression has no family molecule set to measure similarity against. Families are sampled, so similarities are lower bounds.